

Randomized comparison of routine vs highly selective use of Doppler ultrasound and biophysical scoring to investigate high risk pregnancies

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Abstract

Objective—To compare routine versus highly selective use of Doppler ultrasound and biophysical scoring in higher risk pregnancy.

Design—A pragmatic randomized trial.

Setting—St James's University Hospital, Leeds.

Subjects—500 pregnant women at high risk of intrauterine growth retardation or still birth.

Interventions—Regular monitoring with biophysical profile assessment and Doppler velocity waveform recording in umbilical and uteroplacental arteries. Results immediately available to clinicians.

Main outcome measures—Gestational age at delivery, obstetric intervention rates and short-term neonatal morbidity.

Results—Risk factors were distributed very evenly between the 250 patients in the study and control groups respectively. A total of 902 biophysical profile and Doppler assessments were done in the 250 study group patients and only in 12 patients in the control group. In the study group, absent end-diastolic flow was found in only 2.7% of all 902 measurements. A persistently abnormal biophysical score was always associated with absence of end-diastolic flow. The mean gestational age at induction of labour was statistically and clinically similar in the two groups and there was no overall statistically significant difference in intervention rates between the two groups. There was a statistically significant lower frequency of depressed 5-min Apgar scores in the study group. Serious neonatal morbidity was also statistically significantly more common in the control group than in the study group.

Conclusions—The use of Doppler ultrasound in higher risk pregnancies does not lead to an increase in iatrogenic preterm delivery. The total rate of positive tests on Doppler ultrasound is very low and persistently abnormal biophysical scores are unlikely to be found in patients where umbilical end-diastolic blood flow is present. Surrogate measures for fetal damage seem to be improved when clinicians have access to Doppler ultrasound assessments.

Much of obstetric care is based on the premise that high risk pregnancies should be monitored

by antenatal tests of fetal well-being. Inexpensive Doppler ultrasound equipment became

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available 4 years ago and we acquired this apparatus, along with a conventional ultrasound machine suitable for generating biophysical profile scores. Abnormalities in Doppler waveforms of the uteroplacental and fetal umbilical circulations have been reported to identify fetuses at particularly high risk of mortality and serious morbidity (Rochelson *et al.* 1987; McCowan *et al.* 1987). More recently, abnormal umbilical artery waveforms have been shown to be strongly associated with fetal hypoxia and acidosis (Nicolaidis *et al.* 1988; Tyrrell *et al.* 1989). It was postulated that Doppler ultrasound would be a useful addition to our repertoire of tests of antenatal fetal well-being and that timely intervention, on the basis of abnormal Doppler results, might improve obstetrical decision-making (Trudinger *et al.* 1987) and prevent intra-uterine death. In addition, because hypoxic cerebral damage may begin before labour (Symonds 1987), and because intrapartum asphyxia is probably more damaging when superimposed on underlying hypoxia, Doppler assessment may lead to intervention that reduces the risk of fetal brain damage. The hypothesis that Doppler is effective in reducing mortality and major morbidity in high risk pregnancy could only be tested with a massive randomized trial (Lilford 1987; De Bono *et al.* 1990), so we have estimated its effect on the occurrence of low Apgar scores and the need for and duration of neonatal special care.

The trial was designed to measure the effect of routine Doppler and biophysical profile testing for all high risk pregnancies as opposed to very selective use of the technology, that is, we compared the intention to use this form of fetal assessment in all high risk pregnancies with the intention not to use it, unless extreme circumstances supervened and early delivery was contemplated. The protocol for the trial placed particular emphasis on the rates of iatrogenic preterm birth, need for neonatal intensive care, and clinical signs of serious neonatal morbidity (hypoxic-ischaemic encephalopathy and major intraventricular haemorrhage). The study was thus designed to answer the pragmatic question: 'should an obstetric department acquire this technology and the necessary staffing for surveillance of all high risk pregnancies?' rather than the question: 'can this technology ever be of use?.'

Subjects and methods

High risk pregnancies were identified by medical

staff, either at the booking clinic or when a pregnancy complication occurred antenatally. Only patients from the Leeds Eastern District who were booked for antenatal care at St James's at the request of their general practitioner were considered eligible for trial entry; patients referred from other hospitals were offered Doppler and fetal biophysical investigations if these were considered appropriate, but not as part of the trial.

Risk factors were recognized in 15% of all pregnancies fulfilling the above criteria. We deemed a fetus to be at increased risk of being born small for gestational age (SGA) if intra-uterine growth retardation was clinically suspected by a senior obstetrician, or if an ultrasound scan reported a fetal abdominal circumference of ≥ 2 SD below the mean for gestational age. We defined a previous SGA baby as a birthweight $<$ the 10th centile for gestational age, and a previous antepartum haemorrhage (APH) as a haemorrhage that led to an emergency operative delivery. Hypertension was defined as a persistent (>48 h) blood pressure of $>140/90$ mmHg, and proteinuria was considered significant if a semi-quantitative measurement (Esbach's test) indicated an excess of 0.5 g/24 h in the absence of a urinary tract infection. Serious maternal illness included a wide range of diseases that may influence pregnancy adversely, for example chronic respiratory or cardiac disorders or a renal transplant.

We specifically excluded women with diabetes as there is evidence to suggest that an abnormal umbilical artery Doppler waveform may not predict adverse outcome in this condition (Tyrrell 1988; Landon *et al.* 1989). Multiple pregnancies were excluded because these groups were enrolled in a separate study. Smoking, low socioeconomic group and alcohol use were not included as risk factors because although they are associated with poor pregnancy outcome, they would have resulted in almost 70% of our antenatal population being defined as high risk.

All the women in our study were randomized by using a random number sequence contained in sealed, sequentially numbered envelopes. Blocking was not used in the randomization schedule. The randomization was performed by the two ultrasonographers involved in the study, neither of whom knew anything about the patients or was involved in their clinical management. It was possible to account for all envelopes and randomization allocations at trial completion.

Randomization resulted in the formation of a routinely investigated and a selectively investigated group. Women in the routinely investigated group had Doppler scans and biophysical profile testing at 28 weeks gestation, or at the time of presentation if risk factors appeared later than this. Thereafter, they had weekly Doppler and fetal biophysical assessments for 3 weeks, followed by fortnightly examinations until delivery. These investigations were made by the ultrasonographers of the fetal assessment laboratory, or by one of us (ST), and were relayed to the clinicians responsible for the woman's care.

The biophysical profile score was modified from that described by Manning *et al.* (1981) by one of us (HNM) and incorporated an ultrasound assessment of amniotic fluid volume and gross fetal limb/body movements, together with response to an external sound stimulus and a non-stressed cardiotocograph (CTG). The method of scoring is shown in Table 1. A score of 8 or 7 was considered normal and an abnormal biophysical score was a score of ≤ 6 . Ultrasound information on amniotic fluid volume and fetal movements was obtained with an Hitachi EUB340 real time scanner using a 3.5 MHz curvilinear transducer. A Sonicaid EM3R cardiotocograph (CTG) recorded the non stressed CTG. External sound stimulation was produced by placing a sound source of 110 dB at 120 Hz on the maternal abdomen adjacent to the fetal head. The sound stimulus comprised four 1-s bursts at 1 s intervals, repeated 2 min later (similar to that described by Birkenfeld *et al.*

1986). Doppler waveforms of the maternal uteroplacental and fetal umbilical arteries were measured and scored as shown in Table 2. Doppler waveforms were recorded and measured with a continuous wave Doppler instrument linked to a fast fourier spectrum analyser with a 100 Hz 'thump' filter (Doptek 9000, Doptek, Chichester, England). A 4 MHz probe was used throughout. All waveforms were recorded for subsequent reference on standard audio tape. In each woman both right and left maternal uteroplacental waveforms were recorded as described previously (Schulman *et al.* 1986). The fetal umbilical artery is readily identified, although occasionally (for example in severe oligohydramnios), an offset duplex system linking both real time scanner and Doppler instrument was used to pinpoint the vessel. When recording from this artery, care was taken to ensure that the fetus was neither active nor breathing. For each artery a total of five consecutive waveforms were recorded and measured and the mean value was reported. Due to the availability of established normal ranges we used the resistance index of Pourcelot (RI) to assess the uteroplacental artery waveform and the AB ratio to assess the umbilical artery waveform. An abnormal RI was defined as a value of ≥ 0.58 (Campbell *et al.* 1986). We defined an abnormal AB ratio in two grades; the less ominous was a value of >95 th centile for gestational age (Trudinger *et al.* 1985) which scored 1; a more ominous pattern was absence of the end diastolic component and this scored 0 (Table 2).

In practice, the order of assessment was a real-

Table 1. Reporting method for fetal biophysical profile scoring

Biophysical variable	Score
Fetal movement: no. of gross limb or trunk movements in 10 min	
2 movements	2
1 movement	1
No movement	0
Amniotic fluid volume: depth of fluid pool in vertical axis	
>2 cm	2
1–2 cm	1
<1 cm	0
Sound response: fetal movement or FHR acceleration of 15 b.p.m. for 15 s	
Response to both stimulus	2
Response to one stimulus	1
No response	0
Cardiotocography: FHR accelerations of >15 b.p.m. for 15 s within 30 min	
Two reactive episodes	2
One reactive episode	1
No reaction	0

FHR=fetal heart rate, b.p.m.=beats per minute.

Table 2. Reporting method for Doppler waveform scores

Variable	Score
Umbilical artery AB ratio	
<95th centile for gestation	2
≥95th centile, end-diastolic flow present	1
Absence of end-diastolic flow	0
Uteroplacental artery resistance index	
<0.58 on both sides of the uterus	2
≥0.58 on one side of the uterus	1
≥0.58 on both sides of the uterus	0

time scan for amniotic fluid volume estimation and fetal movement count, Doppler measurement, sound stimulus and finally CTG. This proved to be the most effective sequence, as initial identification of the fetal umbilical cord and placental position facilitated (but was not essential to) the remaining investigations and after sound stimulation, fetal movements were commonly prolonged, making amniotic fluid assessment and umbilical artery Doppler measurement more difficult. If an abnormal score was recorded the assessment was repeated at more frequent intervals than given above at the discretion of the obstetrician; this could be as frequently as daily, if requested. The obstetricians were fully aware of our earlier study (Tyrrell *et al.* 1989) that had demonstrated a clear association between abnormal fetal umbilical artery Doppler waveforms, especially those with an absent end diastolic component, and antenatal hypoxia and acidosis.

Obstetric investigation in the selectivity investigated group could include growth scans, CTG and kick charts, all as part of either in-patient or out-patient care. Biochemical fetal assessment is not used at St James's Hospital. Clinicians responsible for the care of women in the selectively investigated arm could only obtain Doppler and biophysical assessments on special request, and this happened in only 12 pregnancies. After delivery, data were collected on the total number of days of antenatal admission, the rate of induction of labour (by any method), mode of delivery (differentiating between elective and emergency caesarean section), 1 and 5-min Apgar scores, birthweight, and admission to the neonatal unit (NNU). For babies admitted to the NNU, detailed information was obtained on the neonatal course. St James's Hospital has a 'transitional' ward for reasonably well low birthweight babies where a high staff to patient ratio

allows for many babies, who would otherwise be admitted to the NNU, to be cared for with their mothers. This system ensures that only babies that truly require the services of an intensive care unit are actually admitted to the NNU.

As far as possible we have used end points that are unambiguous. Clearly the trial is not blinded, the women in the routinely investigated group had the information on Doppler and biophysical variables in their notes available for everyone to see. Women in the selectively investigated group simply had a note on the inner face of the case notes stating 'standard care'. Although the clinician responsible for the decision to deliver by emergency caesarean section was aware of the patient group allocation, we consider it unlikely that the neonatologist's assessment of fetal condition or the midwife's measurement of the Apgar score were biased by this information.

Statistics

Our main objectives were to measure the effect of the technology on obstetric intervention and neonatal morbidity in these high risk pregnancies. If these tests reduced (or increased) the average number of days on the NNU from 0.2 to 0.15 or 0.25 (assumed SD 0.05), 40 patients in each group would be sufficient, rising to 200 to show a difference of 0.2 to 0.18 (power 80%, $P=0.05$). To show a difference in the frequency of low Apgar scores from 12% to 5%, 245 patients in each group would be required to reach a P value of 0.05 with a power value of 80% and smaller changes would be measureable with contingency table analysis. The planned trial of 500 patients was therefore sufficient to measure an appreciable decrease in neonatal morbidity and requirements for neonatal intensive care. The trial would have a 75% chance of showing a reduction in serious neonatal outcome (probable asphyxiated stillbirth, hypoxic/ischaemic encephalopathy and major (grade 3 or 4) intraventricular haemorrhage) at the 10% significance level. The Mann-Whitney U -test was used to compare durations of antenatal admissions, total 5-min Apgar scores, and gestational age at delivery. To compare rates of induction of labour, caesarean section, SGA-infants and a low 5-min Apgar score, a χ^2 -test was used. To examine differences between groups in respect of length of stay NNU and ventilation time, a t -test was used. Odds ratios were

calculated as described by Morris & Garbner (1988).

Results

Comparability of groups (Table 3)

Overall 500 patients were randomly allocated, 250 each to routinely investigated and selectively investigated groups. Some patients were referred with more than one risk factor. There was very little difference in the pattern of risk factors in the two groups. The commonest indications for entry into the study were a diagnosis of intrauterine growth retardation in the current pregnancy, a previous SGA baby, still-birth or neonatal death of a normally formed baby.

Prevalence of abnormal results

A total of 902 biophysical and Doppler assessments were recorded in the 250 patients in the routinely investigated group (mean 3.6, range 1–8) and 580 (65%) showed normal Doppler and biophysical test results. Overall, 886 biophysical assessments were normal (98%) and 12 assessments showed abnormal results both for biophysical variables and Doppler waveforms (either umbilical or utero-placental artery). Four assessments had abnormal biophysical scores but normal Doppler results and all four women had normal biophysical scores on repeat testing the following day. In 60 patients (24%) the umbilical artery waveform had a score of 1 or 0 on at least one occasion and 95 patients (38%) had a score of 1 or 0 in either the umbilical or uteroplacental waveforms or in both on at least one occasion. No patient had a persistently

abnormal biophysical score with a normal umbilical artery Doppler waveform, and only 5.9% of all measurements showed severely abnormal umbilical (no end-diastolic component) or uteroplacental waveforms (both sides RI ≥ 0.58). Regardless of uteroplacental waveform measurements, only 2.7% of all investigations showed a severely abnormal (no end-diastolic component) umbilical artery waveform.

Obstetric interventions (Table 4)

There were no differences between the two groups in the length of antenatal stay or rates of induction of labour and elective caesarean section (the data on duration of antenatal stay and induction of labour were obtained retrospectively, and the case notes could not be traced in 15% of the women). There was no statistically significant difference in the rate of elective delivery (either by caesarean section or by induction of labour). Overall, however, the mean gestational age at delivery was lower in the routinely investigated group than in the selectively investigated group, a statistically significant difference which is explained by the difference in the gestational age at spontaneous onset of labour in the two groups.

The gestational age at which labour was induced was slightly earlier in the routinely investigated group (38.1 weeks) than in the selectively investigated group (38.6 weeks) but this difference was not statistically significant ($P=0.29$, Mann Whitney *U*-test).

Outcome (Table 5)

In the routinely investigated group there were

Table 3. Indication for referral to the trial in the two investigation groups

Indication	Routine (<i>n</i> =250)	Selective (<i>n</i> =250)
Previous stillbirth or neonatal death	35	29
Previous SGA	59	64
Previous APH	1	4
Serious maternal illness	10	14
Maternal age ≥ 40 years	17	14
Hypertension	27	30
Hypertension with proteinuria	7	4
APH in this pregnancy	12	8
IUGR in this pregnancy	90	98

SGA = Small-for-gestational Age;

APH = Antepartum haemorrhage;

IUGR = Intrauterine growth retardation.

Table 4. Obstetric characteristics in the two investigation groups

Variable	Routine	Selective		
No. of patients	250	250		
Total no. of tests*	902	22		
Antenatal stay (mean no. of days/patient)	3.8	3.6	$P=0.41$	
Gestation (weeks)				
At induction of labour Mean (SD)	38.1 (1.8)	38.6 (1.4)	$P=0.29$	
At delivery Mean	37.8	38.3	$P<0.01$	
Obstetric interventions			Odds ratio	(95% CI)
No. of patients†	230	222		
Induction of labour	69	73	0.87	(0.58–1.30)
Elective section	26	20	1.29	(0.70–2.38)
Total elective interventions	92	91	0.96	(0.66–1.40)

*Doppler waveform measurements and biophysical profile assessments.

†The notes could not be retrieved for all 500 patients and these data were not collected prospectively.

247 live and three still births. Two of these stillbirths were due to multiple congenital abnormalities (one of which was detected at the biophysical assessment). One was a fresh stillbirth in the second stage of labour, the baby was SGA and the Doppler waveforms from the umbilical artery had been abnormal for several weeks. Only five babies with a normal umbilical artery waveform were admitted to the NNU. One of these babies was asphyxiated due to a major placental abruption and one had opiate withdrawal. The other three had little morbidity, none required ventilation, there were no intra-ventricular haemorrhages and the longest duration of stay on the NNU was 4 days. A total of 98 babies in the routinely investigated group had a birthweight under the 10th centile (corrected for

gestational age). Doctors frequently appeared to alter their normal clinical management as a result of the assessment in the routinely investigated group. For example caesarean section was carried out within a week in six of the nine patients with an absence of the end diastolic component in the umbilical artery waveform. In the routinely investigated group as a whole, 13 babies requiring admission to the NNU, and all three stillbirths occurred in the pregnancies associated with an abnormal umbilical artery Doppler waveform ($n=41$). A total of nine babies were born before 34 weeks gestation, two after spontaneous labour and seven after obstetric intervention.

There were 12 patients in the selectively investigated group for whom biophysical and Doppler

Table 5. Main and secondary outcome variables in the two investigation groups

Outcome	Routine	Selective	Odds ratio	(95% CI)
Main variables				
Livebirth	247	249	–	–
Stillbirth/NND (abnormal baby)	3 (2)	3 (1)	–	–
Elective delivery <34 weeks	7	6	1.17	(0.38–3.60)
5-min Apgar score <7	3	12	0.24	(0.06–0.86)
NNU admissions	18	19	0.94	(0.48–1.84)
Serious neonatal morbidity	1	8	0.12	(0.02–0.98)
Hypoxic encephalopathy	0	5		
Major IVH	0	2		
NND/stillbirth related to hypoxia	1	1		
Secondary variables				
Ventilation time (days)				
Mean (SD)	2.8 (5.3)	2.3 (2.2)	$P=0.7$	
Duration of stay (days)				
Mean (SD)	26.5 (22.2)	33.6 (21.5)	$P=0.34$	

NND = Neonatal death; NNU = Neonatal unit; IHV = Intraventricular haemorrhage.

measurements were specifically requested; these patients were then monitored in the same manner as those in the routinely investigated group.

There was one stillbirth and two neonatal deaths among the 250 patients in the selectively investigated group. The stillbirth was structurally normal and SGA but died at 38 weeks gestation without obvious cause. One neonatal death was due to Edwards' syndrome and the other was due to severe respiratory distress at 28 weeks gestation, the pregnancy had been complicated by severe pre-eclampsia.

There were no statistically significant differences between the two groups in respect of admissions to the NNU (Table 5), but there was a statistically significant difference in the proportion of babies with a low 5-min Apgar score (<7). In the groups overall there were no differences in ventilation time or in duration of antenatal stay.

It is of interest to note that the five babies born after 37 weeks gestation who required admission to the NNU in this trial were all from the selectively investigated group. These admissions were for hypoxic/ischaemic encephalopathy (defined as babies with a combination of neonatal seizures, reduced muscle tone and recurrent apnoeic attacks), two of these had the additional complication of meconium aspiration. One of these babies had had a subclinical fetomaternal haemorrhage with an umbilical vein haemoglobin at birth of 3 g/dl. Three babies in the routinely investigated group had postnatal cranial ultrasound scans suggestive of intraventricular haemorrhage (IVH) of grade 1–2. These and all other babies in this group admitted to the NNU were well, with no obvious handicap or disability, at 6-month follow up. Three babies in the selectively investigated group also had cranial ultrasound scan evidence of IVH, one of grade 1, one of grade 3 and one of grade 4. The last two babies now appear to have long-term handicap in that one has hydrocephalus and the other is epileptic with developmental delay at 9 months of age. A third baby from the selectively investigated group, who was admitted to the NNU, has developmental delay at 8 months follow-up. At 14-month follow-up, the baby with a cord haemoglobin of 3 g/dl has developmental delay, with a head circumference below the third centile, and spasticity. The remainder of the group that were admitted to the NNU appear well. There were significantly fewer serious perinatal problems attributable to hypoxia in the

routinely investigated group than in the selectively investigated group ($P = 0.05$, Table 5).

Discussion

There were no differences between groups in the pattern of risk factors or in the obstetric management variables examined. We found no increase in potentially serious iatrogenic preterm birth at <34 weeks, and in addition there is no evidence that the rate of caesarean sections has increased. Similarly, both the duration of in-patient antenatal stay and the rate of induction of labour, although high overall (reflecting the high risk study population), are not significantly different between the two groups. We can therefore justifiably argue that although obstetricians were aware of a link between abnormal Doppler and biophysical assessments and hypoxia, use of this technology did not increase the risk of obstetric intervention appreciably, and that the evidence strongly suggests that the use of this technology was not harmful in our hands. Significantly more pregnancies in the selectively investigated group had a low 5-min Apgar score <7 . Our results therefore suggest that routine biophysical testing with Doppler assessment correctly identifies the small minority of fetuses at risk of hypoxia and alerts the obstetrician to allow for a timely delivery. Indeed, as there was no patient with a persistently abnormal biophysical profile in the presence of a normal umbilical artery waveform, we suggest that Doppler assessment alone in a high risk pregnancy is sufficient, with regular biophysical testing only in those pregnancies complicated by an abnormal umbilical artery waveform. This approach appears to allow for the effective use of Doppler; rarely does Doppler information alone direct the clinician to deliver the pregnancy, but it is an indication for more intense monitoring, allowing for timely delivery without excessive intervention.

Observation of the outcome in patients in the routinely investigated group has confirmed that a normal Doppler waveform in the umbilical artery is strongly linked with a normal neonatal outcome, even in babies who are very SGA (below the 3rd centile). It is assumed that these SGA babies are either constitutionally small or small secondary to other extrinsic factors such as maternal smoking rather than inadequate placental blood flow or function. This is supported by the reported association between increasingly abnormal umbilical artery waveforms and an

increasing risk of fetal hypoxia and acidosis (Tyrrell *et al.* 1989).

We now offer Doppler investigation to all high risk pregnancies from 28 weeks gestation as a routine. We have started a randomized trial involving low risk primigravidae where the case for this new technology is not yet made, because the proportion of false positives may be higher, predictive values lower and clinical effect less favourable.

In the meantime, we await the results of other trials of Doppler ultrasound in high-risk pregnancy and look forward to combining those results with ours for meta-analysis (Chalmers *et al.* 1989).

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